Deep Learning-Based Classification of Drinking Water Waste Using Satellite and Visual Data for Sustainable Environmental Design

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SCOPE

The escalating challenges of urbanization, water pollution, and resource inefficiency necessitate advanced technologies for sustainable waste management [1]. Drinking water waste, often overlooked in environmental assessments, poses significant ecological risks due to its complex composition and dispersed origins [2]. This study aims to develop a novel, AI-driven classification framework that integrates remote sensing and image data to identify and categorize drinking water waste from domestic, institutional, and industrial sources [3].

METHODOLOGY

The proposed system combines structured survey data with high-resolution image datasets—including satellite and ground-level visual records—capturing the spatial and material diversity of drinking water waste. A novel Convolutional Neural Network (CNN)-based deep learning architecture is developed to perform multi-class classification of waste types, with an emphasis on materials such as organic residues, residual treatment chemicals, and microplastics. The model is optimized using Bayesian optimization to efficiently navigate the hyperparameter space and enhance predictive accuracy. Training and evaluation are conducted using a curated and annotated dataset representing diverse environmental and lighting conditions. Data preprocessing includes normalization, augmentation, and feature enhancement techniques to improve model generalization.

RESULTS AND DISCUSSION

Preliminary experiments of the proposed CNN architecture are expected to significantly outperform existing models reported in recent literature. The Bayesian optimization framework contributed to reducing training time and improving convergence stability. The novelty of the model lies in its capacity to integrate spectral and spatial features across heterogeneous data sources, enabling fine-grained classification of visually similar waste types. These findings demonstrate the utility of AI-driven systems for environmental diagnostics, offering actionable insights for waste minimization, pollution control, and eco-efficient infrastructure planning.

CONCLUSIONS

This study presents a technically innovative and environmentally impactful approach to the classification of drinking water waste using deep learning and image-based analytics. The proposed CNN model, enhanced through Bayesian hyperparameter optimization, establishes a new performance benchmark in the field. Its application offers a scalable, reproducible solution for integrating AI into environmental monitoring and urban sustainability planning.

REFERENCES

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